

your company as you talk about your work. OpenOffice.org's Impress? KOffice? How about taking a look at something different? You will like the results, I promise.

Beamer (<http://latex-beamer.sourceforge.net/>) is a LaTeX class for making presentation slides. In LaTeX parlance, it is just another Document Class. You shouldn't even attempt to use Beamer if you have no idea what LaTeX is. However, if you already use LaTeX (and chances are, you swear by it), then Beamer is for you. The easiest way to install it is to use the package manager in your Linux distribution. You will see that some more packages that provide other extra LaTeX classes need to be installed. Prominent among these are *latexcolor* and *pgf*.

There is, of course, a way to install the LaTeX class manually. It's easy to do, but not easier than the previous method. The LaTeX Beamer user guide shows you how to do it.

I would like to point to my LFY article "Typesetting Presentations with Beamer", a PDF copy of which you can download from <http://amitskaha.files.wordpress.com/2010/02/02-beamer.pdf> (or <http://goo.gl/q7N9>).

Other tools you may find useful

I'll quickly mention some of the other tools that I find useful for scientific computing:

- **Sage:** Sage (<http://www.sagemath.org/>) is an open source mathematics software system that brings together a huge number of open source mathematical libraries and tools under one common umbrella. In their own words, the Sage community is building a car, and not reinventing the wheel. The Sage tutorial (<http://www.sagemath.org/doc/tutorial/index.html> or <http://goo.gl/CDFG>) is a good place to start exploring it.
- **Xfig:** Xfig (<http://www.xfig.org/>) is a drawing tool that may be used to create, import and export figures in various other formats. FIG is the native format that Xfig uses to save its images. A useful work-flow that I have learnt is to export your plots from Gnuplot as .fig images, open them in Xfig to add the relevant text/labels to your plots, and then export the plot in a format you desire.
- **GNU Scientific Library:** (see <http://www.gnu.org/software/gsl>) This is a C/C++ numerical library providing support for linear algebra, complex numbers, numerical differentiation and others.

The small ones

One category of tools that I haven't talked about yet is some of the Linux utilities shipped as part of the Linux system: shell scripting, *awk* and *sed*. I must confess that I have just a working knowledge of each one, but can get things done with them—which is what matters.

- **Shell scripting:** Very useful for running batch jobs; fire up a job, and it should finish without requiring any intermediate user intervention. It's all the more useful when you have to run an experiment of yours a large

number of times. Writing a shell script is often as simple as just writing several separate commands in a for loop in a script file, and running that file. Of course, that's an oversimplification, but sometimes it is literally that simple to set up a batch job. Do not get put off by the title, but the Advanced Bash-Scripting guide (<http://tldp.org/LDP/abs/html/>) is a good place to get started with scripting for the BASH shell.

- **awk:** The only way I have used *awk* so far (and I often do so) is to extract text from huge files containing numerical and textual data. Even with this limited functionality, it can be a useful tool when you want to extract text that is stored in a tabular format. The GNU Awk User's Guide (http://www.gnu.org/manual/gawk/html_node/index.html or <http://goo.gl/DyBP>) is where you start exploring *awk*.
- **sed:** *sed* is a stream editor, and is useful to carry out batch editing, as well as editing of individual text files. The three-part series on *sed* at <http://www.ibm.com/developerworks/linux/library/l-sed.1.html> (or <http://goo.gl/asab>) is a comprehensive treatment of the features of *sed*. It's often the case that we seem unaware of some great

tool that would simplify our work, as we go about our everyday research. My hope as a novice researcher is to acquaint the reader with the kind of software tools available in the open source world. I have made no claims to this being an exhaustive list. There are a lot of other tools out there, which I haven't used yet, so I have refrained from writing about them. I hope you have enjoyed reading this article as much as I have writing it!

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This article is a reprint, and was first published on linuxgazette.net.

Disclaimer

I wrote this article just after having finished my latest task—conducting numerous experiments that involved processing a lot of data, and finally writing a paper on it—in just under five days. I wonder how I'd do it without some of the tools described in this article, quite apart from the pleasure of working on Linux systems. My views are biased, and are meant to be taken *cum grano salis*, but I hope you will give me a fair hearing. 

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